

PHOTOSYNTHETIC RESPONSES TO LIGHT OF AUSTRALIAN AND CHILEAN SPECIES OF *NOTHOFAGUS* AND THEIR RELEVANCE TO THE RAINFOREST DYNAMICS

BY JENNIFER READ AND ROBERT S. HILL

*Botany Department, University of Tasmania, Box 252C,
GPO Hobart, Australia 7001*

(Accepted 12 August 1985)

SUMMARY

Photosynthetic responses to light have been examined in nine temperate rainforest species, including *Nothofagus* species from Australia and Chile, in order to correlate comparative light requirements with regeneration patterns and forest dynamics. *Nothofagus gunnii* (Hook. f.) Oerst. has the highest light requirements of the Australian *Nothofagus* species and this correlates with its failure to regenerate in closed forest. *N. alpina* (Poepp. et Endl.) Oerst. is the most shade-tolerant of the *Nothofagus* species examined in this study, and this result conforms with regeneration patterns observed in Chilean rainforest. The Australian species, *N. cunninghamii* (Hook.) Oerst. and *N. moorei* (F. Muell.) Krasser, and the Chilean species, *N. obliqua* (Mirbel) Oerst. and *N. dombeyi* (Mirbel) Oerst., have similar photosynthetic characteristics. Consequently, the comparative light requirements of the *Nothofagus* species described here do not explain differences in the dynamics of rainforests dominated by these species. Differences in the regeneration characteristics of co-occurring understorey and shade-tolerant canopy species appear to be more important determinants of the dynamics of these forests.

Key words: Australia, Chile, forest dynamics, *Nothofagus*, photosynthesis.

INTRODUCTION

There are currently 34 described species of *Nothofagus*, distributed through southern South America, New Zealand, Australia, New Caledonia and New Guinea (Hill, 1983a). The species of New Guinea and New Caledonia occur in tropical latitudes and constitute a distinct taxonomic group. The other species are commonly components of cool temperate rainforests, which have floristic and structural similarities throughout the high latitudes.

There are three *Nothofagus* species in Australia. *N. gunnii* (Hook. f.) Oerst., the only deciduous *Nothofagus* species outside South America, is endemic to Tasmania and occurs in a range of forms from small tree to prostrate shrub in small patches in the highlands. It fails to regenerate in closed forest, although seedlings may be common on forest edges (Read & Hill, unpublished data). *N. cunninghamii* (Hook.) Oerst. is widespread in cool temperate rainforest in Victoria and Tasmania, in lowland and montane communities. It regenerates continuously throughout the range of rainforest associations so far studied (Howard, 1981; Read & Hill, 1985, unpublished data). The closely related *N. moorei* (F. Muell) Krasser (Hill, 1983b) has a localized distribution on the highlands of north-eastern New South Wales and south-eastern Queensland. Unlike *N. cunninghamii* associations, forests dominated by *N. moorei* are commonly bordered by rainforests consisting of warm

temperate and sub-tropical species. *N. moorei* is not regenerating continuously on some sites and is being replaced by warm temperate and sub-tropical rainforest species (Turner, 1976; Howard, 1981; Read & Hill, 1985). The differences in dynamics of these forests suggests that there may be differences in light requirements among the Australian *Nothofagus* species, or in the characteristics of co-occurring species.

In Chile and New Zealand, *Nothofagus* species tend to be restricted to sites which are climatically or edaphically sub-optimal, or to sites which are exposed to recurrent catastrophic disturbance (Wardle, 1983, 1984; Veblen *et al.*, 1981). Wardle (1984) suggests that *Nothofagus* is better able to tolerate harsh conditions than most other rainforest species but that its seedlings are poor competitors. This poor competitive ability on the more equable sites (moist and mild on fertile soils) has been related at least in part to higher light requirements of the *Nothofagus* species relative to co-occurring canopy species (Veblen *et al.*, 1981; Wardle, 1970). In south-eastern Australia, where *N. cunninghamii* dominates rainforest on equable sites and regenerates continuously, only *Atherosperma moschatum* Labill. has been shown to be more shade-tolerant (Read, 1985a), and it is limited in its ability to expand its distribution in the rainforest due to regeneration almost entirely by basal sprouts from older stems (Read & Hill, 1985). This paper examines: (1) The comparative light requirements of the three Australian *Nothofagus* species in order to relate these to differences in the dynamics of the rainforests. (2) The light requirements of *N. moorei* relative to co-occurring species (*Ceratopetalum apetalum* D. Don and *Doryphora sassafras* Endl.), in order to relate any differences to the dynamics of the *N. moorei* rainforests. (3) The light requirements of *Eucryphia moorei* F. Muell. relative to co-occurring species (*D. sassafras* and *C. apetalum*) in order to determine whether the relationship between dominant and sub-dominant canopy species in terms of light requirements is similar to the *N. moorei* rainforests. *E. moorei* dominates cool temperature rainforest in north-east Victoria and south-east New South Wales, between the distributions of *N. cunninghamii* and *N. moorei*. (4) The light requirements of *N. cunninghamii* relative to the Chilean lowland and mid-altitude species, *N. dombeyi* (Mirbel) Oerst., *N. alpina* (Poepp. et Endl.) Oerst. and *N. obliqua* (Mirbel) Oerst., in order to determine whether differences in dynamics between the *N. cunninghamii* rainforests of Tasmania and Victoria, and the lowland and mid-altitude *Nothofagus* rainforests of south-central Chile (39 to 42° S) can be related to differences in light requirements of the *Nothofagus* species.

N. cunninghamii is more widespread in Tasmania than Victoria and has a greater diversity of species associations (Howard, 1981). This paper concentrates on the Tasmanian rainforest though discussion of *N. cunninghamii* dynamics applies equally to *N. cunninghamii* associations in Victoria (Read & Hill, 1985).

MATERIALS AND METHODS

The Tasmanian species were grown from seed collected at Mt Field National Park (42° 41' S, 146° 40' E). Seedlings of the New South Wales species were collected from Banda Banda (31° 10' S, 152° 24' E) (*N. moorei*, *C. apetalum* and *D. sassafras*) and from Clyde Mountain (35° 36' S, 149° 57' E) (*E. moorei*). The Chilean species were grown from seed collected from the botanical gardens of the Universidad Austral de Chile, Valdivia. Seedlings of each species (six months to

two years old at the start of the conditioning period) were grown in contrasting light regimes in a glasshouse from August to December 1984. All plants were grown in sandy loam, watered daily and given nutrient weekly. This experiment was performed in conjunction with a comparative study of the light requirements of Tasmanian rainforest canopy species, and methods and results have been described for *N. cunninghamii* and *A. moschatum* already (Read, 1985a). Eight plants of each species were grown unshaded in the glasshouse (approx. 80 % of the PAR of natural sunlight), and the same number shaded from direct sunlight so that maximum photon flux density (PFD) of unfiltered diffuse light was 30 to 80 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$. The daily maximum temperatures in the glasshouse for the seven weeks prior to and during measurement ranged from 23 to 31 °C with a mean maximum daily temperature of 26 °C. Leaf temperature varied by less than 1 °C between the two light regimes.

N. gunnii has a poor survival rate when grown outdoors or in glasshouse conditions in Hobart. This is probably due to climatic differences between Hobart and the areas of natural occurrence of *N. gunnii*. Therefore, comparison of this species with *N. cunninghamii* was made by growing seedlings of both species in a controlled environment growth cabinet at 23 °C day temperature and 20 °C night temperature. The light source was composed of a 400 W metal arc lamp, eight 40 W fluorescent lamps and four 60 W incandescent lamps during a 12 h photoperiod. One group of plants was positioned to receive a PFD of 700 to 800 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$ (daily photon flux = 30 to 35 mol quanta $\text{m}^{-2} \text{d}^{-1}$). This was the highest PFD possible in the growth cabinet. The second group of plants was shaded from the 400 W metal arc lamp and received 50 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$ (daily photon flux = 2 mol quanta $\text{m}^{-2} \text{d}^{-1}$) from the fluorescent and incandescent lights. Leaf temperature was 1 to 2 °C higher in the plants grown under the higher light intensity during the photoperiod. Leaves of *N. gunnii* produced in the shade regime are similar in form to juvenile leaves (described by Hill, 1984) and are not suitable for measurement of photosynthesis because of their small size and arrangement which prevents enclosure in the leaf chamber.

An open gas exchange system incorporating a Series 225 infra-red gas analyzer (Analytical Development Co. Ltd.) was used to measure the net rate of CO_2 assimilation at selected incident photon flux densities. The light source was a Sylvania 400 W metal arc lamp. Photosynthetically active radiation was measured by a LiCor LI-185 meter with quantum sensor. The light intensities were obtained by varying the distance of the leaf chamber from the light source, and using shade cloth screens. Humidified air (317 to 340 $\mu\text{l l}^{-1} \text{CO}_2$) was pumped through a leaf chamber of cross-sectional area 10 cm^2 at a flow rate of 0.4 l min^{-1} . Moisture was condensed from the gas lines prior to analysis. CO_2 differentials were recorded on chart following equilibration of the leaf to the incident light conditions. Responses were first measured at 750 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$ and subsequently at decreasing light intensities. Response at PFDs higher than 750 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$ were measured last to avoid potential lasting effects of photoinhibition at high PFDs. Leaf temperature was monitored by a thermocouple on the lower leaf surface and maintained at 21 ± 1 °C. Temperatures of 20 to 23 °C were optimal for all species. Leaf photosynthesis was measured only on young fully expanded leaves which developed in the experimental light conditions. One leaf per plant was sampled, with five to eight replicates. Leaf area was measured photometrically. Leaf chlorophylls were extracted from one leaf per plant of each species, in 80 % (v/v) acetone and estimated spectrophotometrically.

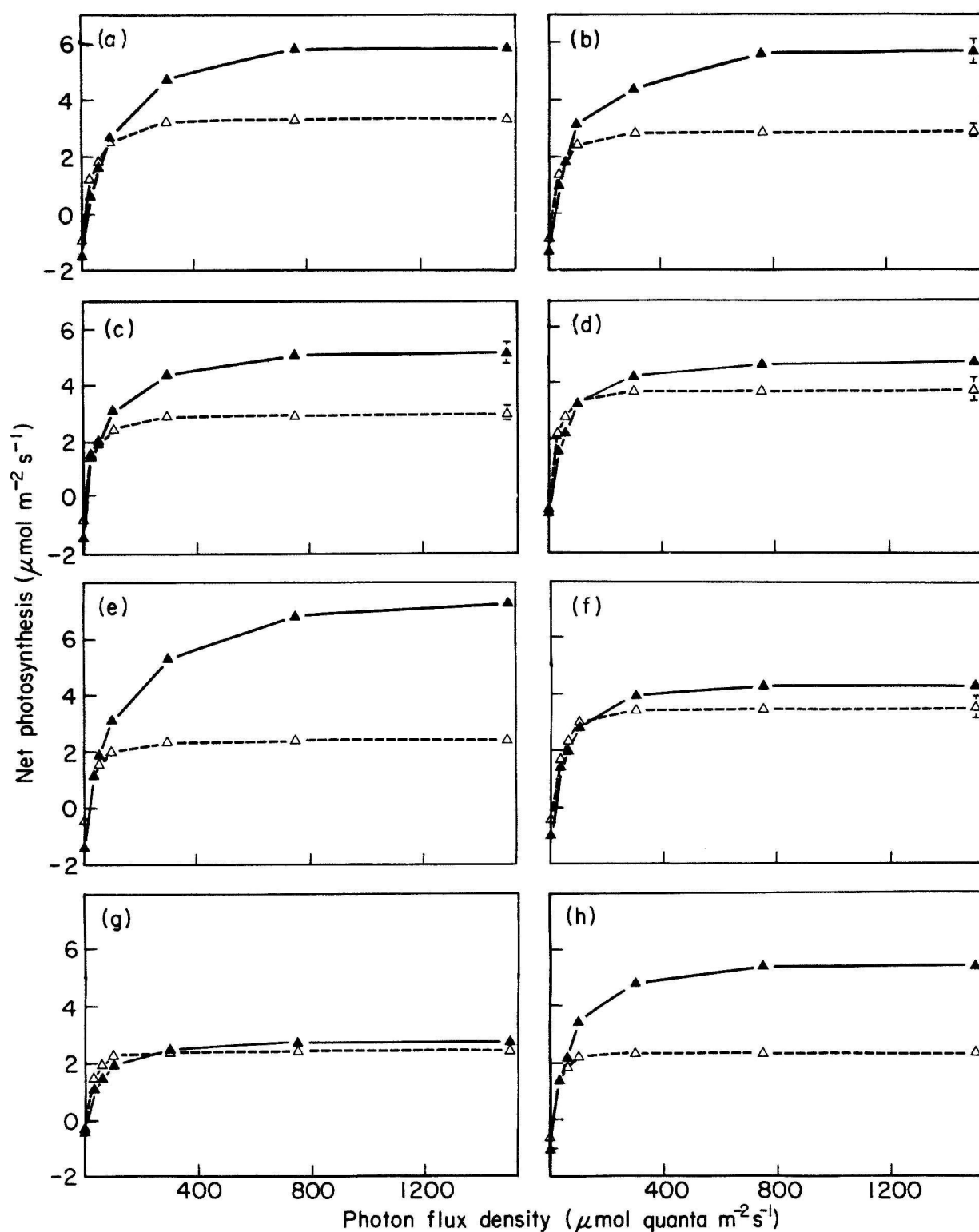


Fig. 1. Light-dependence curves of photosynthesis of rainforest species grown in sun (▲) and shade (△) conditions in a glasshouse. Each point is the mean of five to eight replicates. The largest standard error is shown on each curve, where it is greater than 0.25. (a) *N. cunninghamii* (after Read, 1985a); (b) *N. moorei*; (c) *N. dombeyi*; (d) *N. alpina*; (e) *N. obliqua*; (f) *D. sassafras*; (g) *C. apetalum*; (h) *E. moorei*.

RESULTS

All species show some photosynthetic acclimation to the light regime during growth. *N. moorei*, *N. cunninghamii* and *N. dombeyi* grown in full sunlight have higher dark respiration rates (DRR), instantaneous light compensation points (LCP) and maximum rates of net photosynthesis ($P_{\max}$) on a leaf area basis than shade-raised seedlings (Fig. 1, Tables 1 and 2). Sun-raised seedlings of *N. obliqua* and *E. moorei* have a higher DRR (Table 1) and $P_{\max}$ (Fig. 1) than shade-raised seedlings. Shade-raised seedlings of *N. alpina* have a lower LCP but there is no

Table 1. Leaf characteristics and details from the light-dependence curves of plants raised in sun and shade conditions in the glasshouse

Species	Growth light regime	Specific leaf area (m ² g ⁻¹ × 10 ⁻³)	Chlorophyll (mg m ⁻²)	Dark respiration rate (μmol m ⁻² s ⁻¹)	Light compensation point (μmol quanta m ⁻² s ⁻¹)	P _{max} (μmol g ⁻¹ leaf d. wt s ⁻¹ × 10 ⁻³)
<i>Nothofagus cunninghamii</i> †	Sun	131 ± 7	383 ± 24	1.73 ± 0.12*	27 ± 3*	8.10 ± 0.61*
	Shade	149 ± 8	406 ± 15	1.05 ± 0.12	17 ± 1	4.96 ± 0.31
<i>Nothofagus moorei</i>	Sun	117 ± 3	403 ± 49	1.31 ± 0.15*	22 ± 3*	6.75 ± 0.48*
	Shade	138 ± 12	416 ± 13	0.90 ± 0.14	13 ± 1	4.04 ± 0.58
<i>Nothofagus obliqua</i>	Sun	228 ± 7*	263 ± 25*	1.44 ± 0.26*	19 ± 3	16.70 ± 1.12*
	Shade	442 ± 4	460 ± 11	0.44 ± 0.10	12 ± 2	10.79 ± 1.46
<i>Nothofagus alpina</i>	Sun	238 ± 20*	246 ± 26	0.65 ± 0.11	10 ± 1*	11.18 ± 0.95*
	Shade	417 ± 18	293 ± 10	0.46 ± 0.10	7 ± 0	15.17 ± 1.10
<i>Nothofagus dombeyi</i>	Sun	137 ± 16*	440 ± 41*	1.38 ± 0.15*	20 ± 2*	7.04 ± 1.14
	Shade	316 ± 16	541 ± 8	0.85 ± 0.13	13 ± 1	8.85 ± 1.35
<i>Ceratopetalum apetalum</i>	Sun	113 ± 5*	337 ± 32*	0.40 ± 0.07	9 ± 1	3.15 ± 0.31*
	Shade	220 ± 26	434 ± 22	0.31 ± 0.04	6 ± 1	5.29 ± 0.50
<i>Doryphora sassafras</i>	Sun	109 ± 4*	433 ± 10*	0.99 ± 0.18*	12 ± 3*	4.71 ± 0.36*
	Shade	196 ± 19	611 ± 31	0.37 ± 0.06	6 ± 2	7.11 ± 0.59
<i>Eucryphia moorei</i>	Sun	137 ± 6*	377 ± 20*	1.02 ± 0.07*	14 ± 2	7.26 ± 0.40*
	Shade	186 ± 15	487 ± 1	0.60 ± 0.14	11 ± 2	4.26 ± 0.27

Each value is the mean of five to eight replicates, with standard errors.
* Sun and shade treatments of the same species differ at the 95 % significance level.
† (Read, 1985).

significant difference in DRR (Table 1) or P_{max} on a leaf area basis [Fig. 1(d)]. Shade-raised seedlings of *D. sassafras* have a lower DRR and LCP (Table 1), and a slightly lower P_{max} on a leaf area basis than sun-raised seedlings [Fig. 1(f)]. *C. apetalum* shows no significant difference in DRR, LCP (Table 1) and P_{max} on a leaf area basis [Fig. 1(g)] between treatments. All species have lower light saturation points in shade-raised seedlings (Figs 1 and 2) and there is no significant difference in apparent quantum yield between treatments or species (mean quantum yield is 0.066 μmol CO₂ μmol quanta⁻¹). There is no evidence of photoinhibition at PFD of up to 2500 μmol quanta m⁻² s⁻¹.

There is no significant difference between *N. moorei* and *N. cunninghamii* in DRR, LCP or P_{max} on a leaf area basis or leaf dry weight basis, in either sun- or shade-raised seedlings. Similarly, *N. cunninghamii* and *N. moorei* do not differ from *N. dombeyi* in any of these features except that P_{max} on a leaf dry weight basis is higher in shade-raised *N. dombeyi* than in shade-raised seedlings of the other species, due to the high specific leaf area (SLA) of shade-raised *N. dombeyi*.

N. alpina, *C. apetalum* and *D. sassafras* have lower DRRs and LCPs than the other species (Table 1), and have a relatively poor ability to acclimate photosynthetically to high light intensities during growth (Fig. 1). Shade-raised seedlings of these species have a higher P_{max} on a leaf dry weight basis than sun-raised seedlings due to high SLAs (Table 1).

The three deciduous *Nothofagus* species, *N. gunnii*, *N. alpina* and *N. obliqua*, have a high SLA and P_{max} on a dry weight basis compared with seedlings of the other species (Tables 1 and 2). The P_{max} of *N. obliqua* is highest in sun-raised seedlings and of *N. alpina* is highest in shade-raised seedlings (Table 1). *N. gunnii* has a higher DRR, LCP and light saturation point than *N. cunninghamii* (Fig. 2, Table 2). It also has a high P_{max} on both leaf area and leaf dry weight bases (Fig. 2, Table 2).

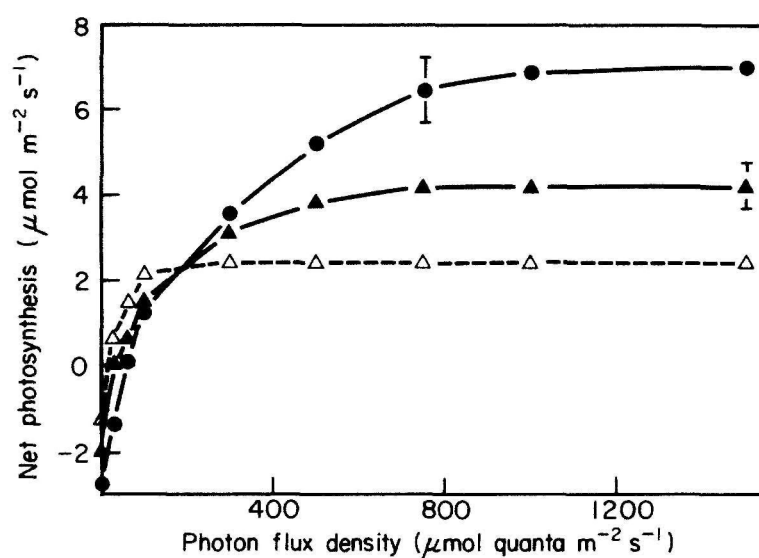


Fig. 2. Light-dependence curves of photosynthesis of seedlings of the Tasmanian *Nothofagus* species grown at high (700 to 800 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$) and low (50 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$) photon flux densities in a controlled environment growth cabinet. Each point is the mean of five replicates. The largest standard error is shown on each curve, where it is greater than 0.25. *N. cunninghamii* raised in high PFD ($\blacktriangle$) and low PFD ($\triangle$). *N. gunnii* raised in high PFD ($\bullet$).

DISCUSSION

Australian *Nothofagus* species

N. gunnii has a higher light requirement for net photosynthesis than *N. cunninghamii*, based on its higher P_{max} , LCP, DRR and light saturation point and its production of juvenile leaves in shaded conditions. This is consistent with its observed failure to regenerate in closed vegetation (Read & Hill, unpublished data) and is in contrast to the continuous regeneration of *N. cunninghamii* in closed forest in which it commonly occurs with the more shade-tolerant *Atherosperma moschatum* (Read & Hill, 1985).

N. cunninghamii and *N. moorei* do not differ significantly in any of the responses to light examined. However, *N. moorei* regenerates in rainforest only in the species-poor forests occurring in the cooler climates at high altitudes and latitudes, although it commonly regenerates from seed throughout its range on disturbed sites and in the more open-canopied *Eucalyptus* forests (Herbert, 1936; Fraser & Vickery, 1938; Turner, 1976; Read & Hill, 1985). Sub-tropical and warm temperate species including *Doryphora sassafras* and *Ceratopetalum apetalum*, commonly occur with *N. moorei* in its warmer climatic range and regenerate continuously (Turner, 1976; Read & Hill, 1985). Similarly, in southern New South Wales *Eucyphia moorei* commonly dominates temperate rainforest at high altitudes and in its lower altitudinal range occurs with *D. sassafras* and *C. apetalum* (Helman, 1983). The low DRR, LCP and poor photosynthetic acclimation to high light conditions exhibited by *D. sassafras* and *C. apetalum* are typical of shade-tolerant plants (Björkman, 1982), indicating that they are more shade-tolerant than both *N. moorei* and *E. moorei*. It appears that the low altitude and latitude limit of *N. moorei* regeneration within undisturbed rainforest is caused primarily by its shade-intolerance relative to co-occurring warm temperate and sub-tropical tree species. This may also be true of *E. moorei* but there is little information regarding its regeneration.

Table 2. Leaf characteristics and details from the light-dependence curves of plants raised in a controlled environment growth cabinet

Species	Growth light regime ($\mu\text{mol quanta m}^{-2} \text{ s}^{-1}$)	Specific leaf area ($\text{m}^2 \text{ g}^{-1} \times 10^{-3}$)	Chlorophyll (mg m^{-2})	Dark respiration rate ($\mu\text{mol m}^{-2} \text{ s}^{-1}$)	Light compensation point ($\mu\text{mol quanta m}^{-2} \text{ s}^{-1}$)	P_{max} ($\mu\text{mol g}^{-1} \text{ leaf d. wt s}^{-1} \times 10^{-4}$)
<i>Nothofagus cunninghamii</i>	700–800	119 ± 9*	402 ± 9*	1.90 ± 0.08*	31 ± 4*	5.06 ± 0.39*
	50	152 ± 11	451 ± 12	1.27 ± 0.18	18 ± 2	3.65 ± 0.21
<i>Nothofagus gunnii</i>	700–800	151 ± 7	380 ± 14	2.78 ± 0.10	62 ± 7	11.97 ± 1.01

Each value is the mean of five replicates, with standard errors.
* Values differ between light treatments at the 95 % significance level.

Table 3. Site data from Nothofagus associations in Chile and Australia

Site	Altitude (m a.s.l.)	Potential diffuse and direct sunlight (%)	No. of canopy species*		Reference
			Total	<i>Nothofagus</i> † Shade-tolerant‡	
Chile§					
Conguillo N.P. 38° 40' S, 71° 39' W	C3 1460	55.8 ± 7.5	2	1 (1)	Veblen (1982)
	4 1440	29.4 ± 5.6	2	1 (1)	
	5 1410	41.0 ± 8.3	2	1 (1)	
	6 1330	37.0 ± 1.7	2	1 (0)	
Nahuelbuta N.P. 37° 38' S, 73° 03' W	N1 1330	28.2 ± 2.5	2	1 (1)	
Antillanca 40° 47' S, 72° 12' W	— 1020	—	2	2 (1)	Veblen <i>et al.</i> (1977a, 1977b)
	— 1040	—	2	2 (1)	
	— 1020	—	2	2 (1)	
	— 1040	—	2	2 (1)	
San Pablo 39° 33' S, 72° 03' W	Sl, 2 810	17.3 ± 1.5	4	1 (0)	Veblen, Schlegel & Escobar (1980)
	3 850	14.0 ± 1.6	4	1 (0)	
Valle Hermoso 40° 01' S, 71° 55' W	V1 830	26.5 ± 2.1	4	2 (0)	
	2 800	22.2 ± 1.3	5	2 (0)	
Pirihueico 39° 54' S, 71° 52' W	3 880	31.3 ± 2.2	4	2 (0)	
	Pl, 2, 3 1060	—	6	2 (0)	Veblen, Ashton & Schlegel (1979)
Lake Villarrica 39° 12' S, 72° 10' W	— 250	—	2	2 (1)	Veblen <i>et al.</i> (1981)
Valle Hermoso 40° 01' S, 71° 55' W	HI 940	46.9 ± 3.0	2	2 (1)	
	2 940	27.9 ± 1.3	3	2 (0)	
	3 950	30.4 ± 1.7	5	2 (0)	
Pilmaiquen 39° 57' S, 71° 55' W	4 880	24.3 ± 1.6	1	1 (0)	
	Q1 740	34.2 ± 1.3	4	2 (0)	
Remeco 39° 46' S, 71° 56' W	2 810	28.0 ± 2.1	2	2 (0)	
	R1 1250	48.2 ± 5.8	2	2 (2)	
Choshuenco 39° 56' S, 72° 03' W	2 1200	46.6 ± 4.5	2	2 (?)	
	3 1250	35.5 ± 2.2	1	1 (0)	
	Cl 1200	40.4 ± 2.5	1	1 (?)	
	2 1120	44.2 ± 3.1	1	1 (?)	

Location	Elevation (m)	Coordinates	Canopy cover (%)	Number of trees	Emergent species	Notes
Banda Banda	—	31° 10' S, 152° 24' E	8.7 ± 1.4	4	1 (0)	Read & Hill (1985)
New England N.P.	—	30° 30' S, 152° 25' E	17.5 ± 3.3	3	1 (1)	
Tasmania	—	—	—	—	—	
Sumac Road	180	41° 09' S, 145° 01' E	24.8 ± 3.5	3	1 (1)	Read (1985b)
Parrawe	580	41° 21' S, 145° 34' E	19.2 ± 4.0	2	1 (1)	
Stephens Rivulet	140	41° 10' S, 144° 57' E	31.3 ± 3.6	2	1 (1)	
Frodsham's Pass	580	42° 50' S, 146° 23' E	12.8 ± 2.8	3	1 (1)	Read (1985b)
Arve Loop	360	43° 08' S, 146° 45' E	22.5 ± 4.3	3	1 (1)	
Mt Michael	740	41° 10' S, 148° 00' E	14.0 ± 2.4	3	1 (1)	
Diddieum	820	41° 22' S, 147° 31' E	14.1 ± 2.6	2	1 (?)	Read (1985b)
Weindorfer's Forest	920	41° 38' S, 145° 57' E	17.0 ± 2.2	2	1 (1)	
Ballroom Forest	940	41° 40' S, 145° 57' E	5.0 ± 1.6	3	1 (1)	
Mt Anne	850	42° 56' S, 146° 26' E	22.4 ± 3.6	3	1 (1)	Read (1985b)
Dempster	220	41° 12' S, 145° 03' E	15.7 ± 2.7	5	1 (1)	
Pieman Dam Road	200	41° 44' S, 145° 26' E	7.3 ± 2.6	5	1 (1)	
Murchison	300	41° 47' S, 145° 38' E	10.5 ± 4.2	6	1 (1)	Read (1985b)
Hastings	120	43° 24' S, 146° 51' E	9.1 ± 3.7	6	1 (1)	

* Species are only included where they form part of the main canopy or are emergent.

† The number in parentheses indicates the number of *Nothofagus* species on the study site that show clearly continuous regeneration.

‡ Species listed are considered to be more shade-tolerant than co-occurring *Nothofagus* species.

§ Chilean canopy trees suggested as more shade-tolerant than *Nothofagus* species: *Aextoxicon punctatum*, *Laurelia philippiana*, *L. sempervirens*, *Dasyphyllum diacanthoides*, *Saxegothea conspicua* and *Persea lingue* (Veblen & Ashton, 1978); *Araucaria araucana* (Veblen, 1982); *Eucryphia cordifolia* and *Weinmannia trichosperma* are uncertain (Veblen *et al.*, 1981) and are not included.

|| *C. apetalum* and *D. sassafras* are the only canopy species on these sites that are more shade-tolerant than *N. moorei*.

¶ *A. moschatum* is known to be more shade-tolerant than *N. cunninghamii* (Read, 1985a). *Anodopetalum biglandulosum* occurs in the canopy with *N. cunninghamii* on some sites and regenerates continuously, but its shade-tolerance relative to *N. cunninghamii* is not known.

Chilean Nothofagus species

The photosynthetic light-dependence curves indicate that the deciduous *N. alpina* is the most shade-tolerant of the *Nothofagus* species examined, based on its low DRR and instantaneous LCP. Veblen, Schlegel & Escobar (1980) note the apparent shade-tolerance of *N. alpina* compared with *N. dombeyi*, based on its more frequent regeneration in canopy gaps in mixed rainforest though the regeneration is commonly insufficient to maintain its abundance. *N. obliqua* does not show any significant difference in photosynthetic characteristics from *N. dombeyi* other than in a higher $P_{\max}$ in sun-grown plants. Both species fail to regenerate in mixed rainforests at low to mid-altitudes in Chile (Veblen, Ashton & Schlegel, 1979).

Comparisons of the Nothofagus forests of south-central Chile and Australia

The structure of the mixed rainforests (rainforests composed of *Nothofagus* and other canopy species) in Chile and Australia is similar in the relatively open understorey and closed canopy, usually with the crowns of the large, old *Nothofagus* trees extending above the other canopy species (described by Howard, 1981 and Veblen *et al.*, 1980). Two hypotheses, that there are physiological differences among these *Nothofagus* species, and that competition with co-occurring species accounts for the differences in dynamics, are proposed by Read & Hill (1985) to account for the differences in dynamics between the *Nothofagus* rainforests of south-central Chile and south-east Australia (dominated by *N. cunninghamii*). These hypotheses can be extended to include *N. moorei* rainforests since *N. moorei* regenerates continuously only at the cooler extremes of its range (high altitudes and/or latitudes) (Howard, 1981; Read & Hill, 1985) and therefore more closely resembles the dynamics of Chilean *Nothofagus* rainforests than the *N. cunninghamii* rainforests.

Shade-tolerance is the most likely physiological characteristic of the *Nothofagus* species to correlate with the differences in dynamics. However results presented here show no significant difference in the photosynthetic light-dependence curves of *N. cunninghamii*, *N. moorei* and *N. dombeyi* that could account for the ability of *N. cunninghamii* to regenerate continuously in mixed rainforest. The levels of canopy cover in Tasmanian rainforests are within the range of those of Chilean and N.S.W. forests (Table 3). The Banda Banda forest has the lowest percentage of canopy opening of the N.S.W. rainforests that were measured (Table 3), and is the site where shade-tolerant trees form a significant proportion of the canopy and where *N. moorei* is not regenerating continuously (Read & Hill, 1985). However, several Tasmanian sites where *N. cunninghamii* is regenerating continuously have similarly low levels of canopy opening (Table 3). Differences in the light climate of these regions due to features such as cloud cover are not known.

The major differences between cool temperate rainforests of Chile and Tasmania are the consistently low number of shade-tolerant canopy species in Tasmania (Table 3) and the light requirements of species which occupy canopy gaps. Gaps in Chilean mixed rainforest are quickly occupied by bamboo (*Chusquea* species) and root suckers and seedlings of shade-tolerant canopy species which may already be present but suppressed and which prevent regeneration of *Nothofagus* species (Veblen *et al.* 1980; Veblen *et al.*, 1981). In contrast, canopy gaps in mixed rainforest in Tasmania are usually occupied by saplings of the relatively shade-intolerant species (e.g. *N. cunninghamii* and *Eucryphia lucida*) except where gaps are created by the death of stems of *A. moschatum* which is usually multi-stemmed

and self-replacing (Read & Hill, 1985; Read, 1985a). *A. moschatum* is the only common canopy species occurring with *N. cunninghamii* that is more shade-tolerant and its distribution through the forest is limited by its localized vegetative reproduction and infrequent establishment from seed (Read & Hill, 1985; Read, 1985a). There is no evidence of any widespread cool temperate shade-tolerant canopy species occurring with *N. moorei*, but in its warmer climatic range it is being replaced by warm temperate and sub-tropical species (Turner, 1976; Howard, 1981) including the relatively shade-tolerant *D. sassafras* and *C. apetalum* (Read & Hill, 1985). The success of these species in autogenically replacing *N. moorei*, compared with the failure of *A. moschatum* to replace *N. cunninghamii*, may result from more successful establishment strategies than *A. moschatum*. The photosynthetic light-dependence curves presented here and by Read (1985a) do not indicate any superiority in the shade-tolerance of *D. sassafras* and *C. apetalum* compared with *A. moschatum*, but ecologically significant differences may not be distinguishable experimentally with such low light compensation points (5 to 6 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$).

The impact of catastrophic disturbance on Nothofagus in Australia.

The succession towards dominance by shade-tolerant species is often interrupted in Chile by recurrent catastrophic disturbance followed by regeneration of *Nothofagus* species (Veblen *et al.*, 1981). The effect of recurrent disturbance on *Nothofagus* in Australia is less certain. Fire is the most commonly reported type of disturbance in temperate Australia. *N. gunnii* is very fire-sensitive and regenerates after fire only in a restricted range of conditions (Kirkpatrick, 1977). *N. cunninghamii* may regenerate after catastrophic disturbance such as fire if the fire-free intervals are long and a seed source is available (Gilbert, 1959). However, unlike the situation in Chile (Veblen *et al.*, 1981), catastrophic disturbance (usually fire) has in general restricted the distribution of *N. cunninghamii* within its climatic range. Communities of plants with relatively xeromorphic characters are promoted by interactions of fire-frequency with vegetation and soil nutrients (Jackson, 1968; Howard, 1981). *N. moorei* forests are often similarly constrained in their distribution (Howard, 1981). The frequency and effect of catastrophic disturbance on the regeneration of *N. moorei* where it occurs with warm temperate and sub-tropical rainforest species would be of particular interest in understanding whether the commonly observed failure of *N. moorei* to regenerate at lower altitudes and latitudes is due to long-term climatic trends (Turner, 1976) or to cycles of infrequent stand regeneration following catastrophic events such as fire or storm.

ACKNOWLEDGEMENTS

We are indebted to Dr F. M. Schlegel for his time and discussion during the visit of J. R. to Chile and for providing seed of the Chilean species used in experiment. J. R. is supported by a Commonwealth Forestry Post-Graduate Research Award.

REFERENCES

- BJÖRKMAN, O. (1981). Responses to different quantum flux densities. In: *Physiological Plant Ecology*, vol. 1: *Physical Environment* (Ed. by O. L. Lange, P. S. Nobel, C. B. Osmond & H. Ziegler), pp. 58–107. Springer-Verlag, Berlin.
- FRASER, L. & VICKERY, J. W. (1938). The ecology of the upper Williams River and Barrington Tops districts. II. The rainforest formations. *Proceedings of the Linnean Society of New South Wales*, **63**, 139–184.

- GILBERT, J. M. (1959). Forest succession in the Florentine Valley, Tasmania. *Papers and Proceedings of the Royal Society of Tasmania*, **93**, 129–151.
- HELMAN, C. E. (1983). An altitudinal gradient in southern New South Wales rainforests. *Proceedings of the Ecological Society of Australia*, **12**, 166–167.
- HERBERT, A. (1936). An advancing antarctic beech forest. *Queensland Naturalist*, **10**, 8–10.
- HILL, R. S. (1983a). *Nothofagus* macrofossils from the Tertiary of Tasmania. *Alcheringa*, **7**, 169–183.
- HILL, R. S. (1983b). Evolution of *Nothofagus cunninghamii* and its relationship to *N. moorei* as inferred from Tasmanian macrofossils. *Australian Journal of Botany*, **31**, 453–465.
- HILL, R. S. (1984). Tertiary *Nothofagus* macrofossils from Cethana, Tasmania. *Alcheringa*, **8**, 81–86.
- HOWARD, T. M. (1981). Southern closed forests. In: *Australian Vegetation* (Ed. by R. M. Groves), pp. 102–120. Cambridge University Press, London.
- JACKSON, W. D. (1968). Fire, air, water and earth – an elemental ecology of Tasmania. *Proceedings of the Ecological Society of Australia*, **3**, 9–16.
- KIRKPATRICK, J. B. (1977). Native vegetation of the west coast region of Tasmania. In: *Landscape and Man* (Ed. by M. R. Banks & J. B. Kirkpatrick), pp. 55–80. Royal Society of Tasmania, Hobart.
- READ, J. (1985a). Photosynthetic and growth responses to different light regimes of the major canopy species of Tasmanian cool temperature rainforest. *Australian Journal of Ecology*, **10**, (In press.)
- READ, J. (1985b). *The Ecology and Dynamics of Rainforest in Tasmania*. Ph.D. Thesis, University of Tasmania.
- READ, J. & HILL, R. S. (1985). Dynamic of *Nothofagus*-dominated rainforest on mainland Australia and lowland Tasmania. *Vegetation*, **63**, 67–78.
- TURNER, J. C. (1976). An altitudinal transect in rain forest in the Barrington Tops area, New South Wales. *Australian Journal of Ecology*, **1**, 155–174.
- VEBLER, T. T. (1982). Regeneration patterns in *Araucaria araucana* forests in Chile. *Journal of Biogeography*, **9**, 11–28.
- VEBLER, T. T. & ASHTON, D. H. (1978). Catastrophic influences on the vegetation of the Valdivian Andes, Chile. *Vegetatio*, **36**, 147–167.
- VEBLER, T. T., ASHTON, D. H. & SCHLEGEL, F. M. (1979). Tree regeneration strategies in a lowland *Nothofagus*-dominated forest in south-central Chile. *Journal of Biogeography*, **6**, 329–340.
- VEBLER, T. T., ASHTON, D. H., SCHLEGEL, F. M. & VEBLER, A. T. (1977a). Distribution and dominance of species in the understorey of a mixed evergreen-deciduous *Nothofagus* forest in south-central Chile. *Journal of Ecology*, **65**, 815–830.
- VEBLER, T. T., ASHTON, D. H., SCHLEGEL, F. M. & VEBLER, A. T. (1977b). Plant succession in a timberline depressed by volcanism in south-central Chile. *Journal of Biogeography*, **4**, 275–294.
- VEBLER, T. T., DONOSO, Z. C., SCHLEGEL, F. M. & ESCOBAR, R. B. (1981). Forest dynamics in south-central Chile. *Journal of Biogeography*, **8**, 211–247.
- VEBLER, T. T., SCHLEGEL, F. M. & ESCOBAR, R. B. (1980). Structure and dynamics of old-growth *Nothofagus* forests in the Valdivian Andes, Chile. *Journal of Ecology*, **68**, 1–31.
- WARDLE, J. A. (1970). The ecology of *Nothofagus solandri*. 4. Growth, and general discussion to parts 1 to 4. *New Zealand Journal of Botany*, **8**, 609–646.
- WARDLE, J. A. (1984). *The New Zealand Beeches: Ecology, Utilisation and Management*. New Zealand Forest Service, Wellington.
- WARDLE, P. (1983). Temperate broad-leaved evergreen forests of New Zealand. In: *Temperate Broad-Leaved Evergreen Forests* (Ed. by J. Ovington), pp. 33–72. Elsevier, Amsterdam.

This document is a scanned copy of a printed document. No warranty is given about the accuracy of the copy. Users should refer to the original published version of the material.